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RESEARCH ARTICLE

Pressureless sintering and properties of additively manufactured $\text{ZrB}_2\text{-SiC}$

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Editor's Choice

The Editor-in-Chief recommends this outstanding article.

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Abstract

Densification behavior, grain growth kinetics, mechanical and thermal properties of pressurelessly sintered zirconium diboride–silicon carbide ($\text{ZrB}_2\text{-SiC}$) fabricated by extrusion of high solids loading pastes have been studied. The effects of sintering temperature (1600–2300°C), heating rate during constant heating rate sintering (10–50°C/min), and time during isothermal sintering (3–24 h) on density and grain size were examined. The apparent activation energy for densification was determined to be 940 ± 90 kJ/mol, suggesting lattice diffusion as the dominant sintering mechanism. Grain growth of both ZrB_2 and SiC followed a normal grain growth power law with grain growth exponent $n = 2$ indicating diffusion-controlled coarsening. The activation energies for grain growth were 267 ± 10 kJ/mol for ZrB_2 and 264 ± 15 kJ/mol for SiC. Sintered microstructures were homogeneous with relative densities of 95%–99% depending on heat treatment schedules. Flexural strength (270 ± 40 MPa) was lower than hot-pressed materials of equivalent composition by a factor of two or more due to the presence of pores originating from paste inhomogeneities. Young's modulus (482 ± 20 GPa) and Vicker's microhardness (22 ± 5 MPa HV0.2), as well as thermal diffusivity and thermal conductivity up to 1800°C were found to be comparable to hot-pressed materials.

KEYWORDS

densification, grain growth, mechanical properties, microstructure, sintering, thermal properties

1 | INTRODUCTION

Zirconium diboride–silicon carbide ($\text{ZrB}_2\text{-SiC}$) ceramics can exhibit flexural strengths >1000 MPa,^{1–3} thermal conductivities >50 W/m K,⁴ and excellent resistance to oxidation and ablation at temperatures above 1500°C,^{5–12} making materials based on this composition attractive for use in high-temperature oxidizing environments. Ceramic additive manufacturing (AM) techniques are being developed to produce ultrahigh-temperature ceramics^{13–16} and

ZrB_2 -based composites^{17–23} with complex geometries for aerospace, energy, and industrial process applications that would otherwise be difficult to fabricate by conventional hot-pressing or casting methods. Compositions featuring constituents with melting points below 1855°C (e.g., Zr²² or ZrSi_2 ²³) have been manufactured using selective-laser melting²² and electron beam melting²³ AM techniques; however, such materials are inherently limited by their maximum service temperature. $\text{ZrB}_2\text{-SiC}$ composites do not possess major constituents having practical melting

points, therefore ceramic AM of such compositions typically employ extrusion of pastes (also referred to as inks) as a green body forming method. Extrusion-based ceramic AM forms green bodies via layer-by-layer deposition of high solids loading paste suspensions whose rheological parameters are controlled by additions of organic binders and dispersants.^{24–26} To minimize differential shrinkage during drying, debinding, and sintering stages that may lead to warpage or cracking, ceramic AM green bodies with relative green densities >50% and homogeneous microstructures are generally desirable. Few studies describing extrusion-based AM of ZrB₂–SiC-based composites densified by sintering or sintering in combination with infiltration approaches have been reported.^{17–21} Reported sintered relative densities and flexural strengths of AM ZrB₂–SiC materials range from 88.84% and 51.9 MPa (flaw limited by average pore size of 588 μm)¹⁷ to 83.55% and 103.5 MPa.²¹ The incorporation of up to 30 vol.% short carbon fiber reinforcements has been reported to significantly increase density (up to 93.09%) and flexural strength (up to 275 MPa) as compared to monolithic AM ZrB₂–SiC.²¹

The performance of monolithic ZrB₂–SiC ceramics is dependent on sintered density and grain size, with higher densities and smaller grain sizes associated with enhanced thermomechanical properties and oxidation resistance.^{2,7,27–33} Design of heat treatment schedules to attain desired microstructures in ZrB₂–SiC requires knowledge of mechanisms and activation energies for densification and grain growth. However, systematic experimentation to determine the activation energy for pressureless sintering of ZrB₂–SiC has not been previously reported. The objective of this study is to investigate the microstructure evolution and mechanical properties of ZrB₂–SiC composites made by pressureless sintering of cast billets from high solids loading paste suspensions suitable for AM.

2 | METHODOLOGY

2.1 | Specimen preparation

ZrB₂ powder (Höganäs Grade B, ABCR GmbH) with 1.5–3.0 μm average particle size and SiC powder (Kyocera Grade UF-250, ABCR GmbH) with 0.65 μm average particle size and specific surface area of 26 m²/g were used as-received. Powders were blended at 70 vol.% ZrB₂ + 30 vol.% SiC (denoted herein simply as ZrB₂–SiC) followed by attrition milling (Model HD-01, Union Process) for 2 h in acetone using silicon carbide media (diameter 5.5–6 mm, Stanford Advanced Materials) and subsequently dried by rotary evaporation (R-124, BUCHI Corporation) at 70°C under vacuum. Dried milled powders were combined

with distilled water and dispersant (Darvan 821A, Vanderbilt Minerals) to produce a slurry which was homogenized by ball milling for 24 h and subsequently de-aired using an electromagnetic jogger (J-1-B, Syntron Material Handling LLC), connected to a vacuum pump for 15 min. Hydroxypropyl methylcellulose binder (METHOCEL J12MS, Dow Chemical Co.) was added to the slurry by high shear mixing (ARE-310, THINKY U.S.A. Inc.) at 2000 RPM until uniform consistency was attained. Resultant pastes were loaded into syringes and extrusion cast into molds to form 2 cm diameter × 0.5–0.8 cm thick billets or connected to a three-axis Cartesian robot for AM of 5 × 5 × 1 cm billets via Ceramic On-Demand Extrusion method^{24–26} for mechanical and thermal properties testing. Details on printing procedures are described in ref.¹⁷

Extrusion cast and additively manufactured billets were dried in a humidity-controlled chamber (MCBH-1.2, Cincinnati Sub Zero) at 10°C and 50% relative humidity for 24–48 h. Binder burnout was conducted in a retort furnace under flowing argon, heating to temperatures of 100, 225, 275, 310, 370, 500, and 600°C at 5°C/min with 1-h holds at each stage to manage the loss of water and volatile organic compounds. Pressureless sintering was performed in a graphite furnace (Model 1000-4560-FP30, Thermal Technology LLC) by heating to 1450 and 1650°C at 20°C/min under vacuum and holding for 1 h at each temperature to facilitate the removal of surface oxides. Following the hold at 1650°C, the furnace atmosphere was switched to flowing argon and the temperature was raised to maximum sintering temperatures from 1600 to 2300°C in increments of 100°C at constant heating rates of 10, 20, 35, and 50°C/min for each maximum sintering temperature. After reaching the target maximum sintering temperature, the furnace was rapidly cooled to below 1600°C at which point no further microstructural changes were expected.

2.2 | Microstructure characterization and sintering analyses

Relative sintered densities were determined by dividing measured bulk densities by the theoretical density of ZrB₂–30 vol.% SiC (5.23 g/cm³) calculated assuming a volume-weighted average density (i.e., rule of mixtures) of constituents ZrB₂ (6.09 g/cm³) and SiC (3.21 g/cm³), neglecting phase impurities. Bulk densities of specimens were determined using the Archimedes method according to ASTM B962-17.

Microstructure analysis was performed by scanning electron microscopy (SEM) in backscattered electron (BSE) mode (Prisma E SEM equipped with energy-dispersive X-ray spectroscopy [EDS], Thermo Fisher Scientific Inc.). Grain sizes of equiaxed ZrB₂ and equivalent

diameters of nonequiaxed SiC grains were determined from high magnification SEM images by manual tracing of grains across a minimum of three different fields of view for each material condition using GIMP and Image J software packages. A minimum of 300 grains per field of view were measured for each cross-section. Herein, the equivalent diameter of a SiC grain is given as the diameter of a circle having the same area as its apparent elliptical cross-section based on measured major and minor axes lengths. The determination of phase compositions was conducted using X-ray diffraction (XRD) analysis on a Bragg–Brentano diffractometer (Philips MPD-XPRT, Malvern Panalytical), employing Cu-K α radiation. Measurements spanned a 2θ range from 5 to 90° with a step size of 0.026°. Grain growth experiments were conducted by applying coarsening heat treatments at 1900–2300°C in increments of 100°C and for durations ranging from 3 to 24 h. These experiments were carried out in an argon atmosphere with heating and cooling rates set at 60°C/min. Initial samples ($\rho = 97 \pm 2\%$) were previously sintered at a maximum temperature of 2200°C using heating and cooling rates of 35°C/min without any holding times. Following each coarsening experiment, samples were cleaned via ultrasonication in acetone and ethanol for 30 min, dried, embedded in epoxy, and polished to a 0.25 μm diamond finish. To reveal the grain boundaries for image analysis, polished samples were mechanically removed from epoxy and thermally etched under empirically determined conditions ranging from 1750 to 1900°C for various times depending on grain size. Smaller grain sizes produced by lower coarsening temperatures and shorter heat treatment durations necessitated higher etching temperatures and longer durations for effective grain boundary visualization.

The kinetics of grain growth were evaluated by use of a power law relationship:

$$d^n - d_0^n = k_0 \cdot e^{-Q/RT} \cdot t \quad (1)$$

where d_0 is the initial grain size, n is the grain growth exponent (typically between 2 and 4), and $k = k_0 \cdot e^{-Q/RT}$ is a thermally activated growth factor with activation energy Q and temperature-independent pre-exponential constant k_0 .

Densification curves (i.e., relative density ρ vs. time at constant heating rate) were fit to a general four-parameter sigmoid function:

$$\rho = C_1 + C_2 / (1 + e^{C_3[t-C_4]}) \quad (2)$$

An expression for the instantaneous densification rate $d\rho/dt$ measured at constant heating rate dT/dt as derived

by Wang and Raj in ref.³⁴ can be written as:

$$\frac{d\rho}{dt} = \frac{d\rho}{dT} \cdot \frac{dT}{dt} = A \cdot \frac{e^{(-Q/RT)}}{T} \cdot \frac{f(\rho)}{d^n} \quad (3)$$

where Q is the activation energy for densification, R is the universal gas constant, T is the absolute temperature, $f(\rho)$ is a function depending only on density, d is the grain size, and A is a material parameter insensitive to d , T , and ρ . Grain growth exponent n identifies a power law dependence on densification with grain size. Linearizing Equation (3) by taking the natural logarithm:

$$\ln \left(T \cdot \frac{dT}{dt} \cdot \frac{d\rho}{dT} \right) = -\frac{Q}{RT} + \ln[f(\rho)] + \ln A - n \ln d \quad (4)$$

The activation energy of densification is calculated by plotting $\ln(T \cdot \frac{dT}{dt} \cdot \frac{d\rho}{dT})$ evaluated at constant density values versus $\frac{1}{T}$ for different heating rates to obtain lines of slope $-Q/R$.

To complement this analysis, a master sintering curve (MSC) approach to analyze densification curves was also employed in accordance with procedures described in ref.³⁵ Briefly, the MSC approach defines a “work of sintering” parameter $\Theta = \int_0^t \frac{1}{T} e^{-Q/RT} dt$ to describe the sintering behavior of arbitrary time–temperature profiles by equating this parameter to an expression $\frac{k_B}{\gamma \Omega D_0} \int_{\rho_0}^{\rho} \frac{G^n}{3\rho\Gamma} d\rho$ that collects the microstructural parameters of surface energy γ , molar volume Ω , grain size G , density ρ , and a geometric scaling factor Γ (k_B is Boltzmann’s constant and D_0 is the pre-exponential for the Arrhenius expression for diffusion coefficient given by $D = D_0 e^{-Q/RT}$, where Q is activation energy for densification). This approach assumes that these microstructural parameters are independent of time and temperature so that the “work of sintering” parameter captures this dependence. To obtain a best estimate for Q using the MSC approach, a least squares regression fitting procedure is employed to minimize the mean residual square error between measured densification data and MSC predicted values using Q as a fitting parameter. MSC trajectories converge onto a single curve when the best-fit Q value adequately describes densification behavior across the full range of experimental parameters.

2.3 | Mechanical and thermal properties testing

Flexural strength at room temperature was determined by a four-point bending test according to ASTM C1161.

Type A specimens were tested using a screw-driven frame (Model 5881, Instron). Reported strength values represent averages from 10 or more specimens. The elastic modulus was assessed using the impulse excitation technique according to ASTM C1259 averaged from at least 10 measurements. Vickers hardness measurements were conducted on the polished surfaces of bar fragments postfracture testing, with a series of at least 10 indents made under various loads and maintained for a duration of 15 s each.

Thermal diffusivity was measured using the laser flash method by ASTM E1461 (Flashline 5000, Anter Corp). Thermal conductivity κ of ZrB_2 -SiC test specimens was calculated by:

$$\kappa = D \cdot \rho \cdot C_p \quad (5)$$

where D is the measured thermal diffusivity, ρ is the density, and C_p is the specific heat capacity. Temperature dependencies for C_p and ρ were estimated to first approximation using tabulated reference data³⁶ and applying a volume-weighted average for composite materials (i.e., rule of mixtures).

3 | RESULTS AND DISCUSSIONS

3.1 | Densification behavior

The XRD pattern of additively manufactured ZrB_2 -30 vol.% SiC is shown in Figure 1. The effect of heating

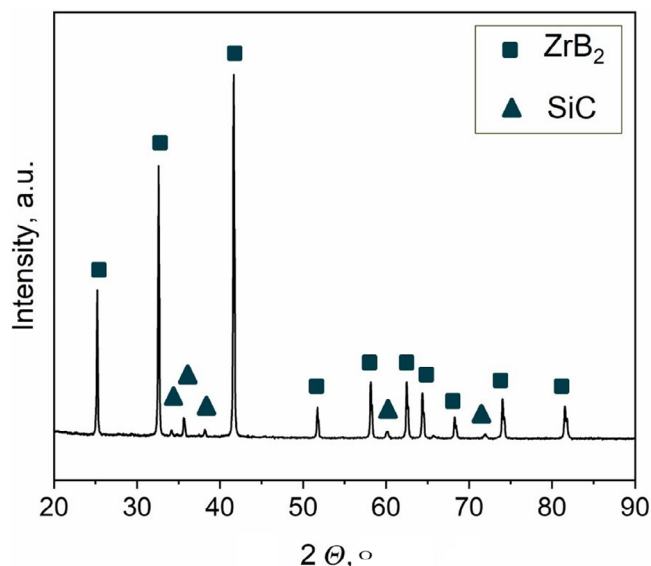


FIGURE 1 X-ray diffraction (XRD) pattern and phase identification of additively manufactured ZrB_2 -SiC. ZrB_2 and SiC indexed peak markers correspond to Powder Diffraction File cards 01-075-1050 and 01-073-1708 (3C-SiC), respectively.

rate on ZrB_2 -SiC sintered density and microstructure has been examined by heating debinded extrusion cast billets to a maximum sintering temperature of 2300°C at four constant heating rates (5, 10, 35, and 50°C/min). Representative SEM images illustrating associated microstructures are shown in Figure 2. Table 1 summarizes grain size and density measurements.

All sintering trials produced uniform two-phase microstructures with homogeneous distributions of SiC within ZrB_2 matrices. Faster heating rates resulted in lower sintered densities and smaller grain sizes due to decreased sintering times at all temperatures. The maximum attained density was $5.19 \pm 0.02 \text{ g/cm}^3$ (99% theoretical) obtained by heating ZrB_2 -SiC at 10°C/min to 2300°C without isothermal holds, which resulted in average grain sizes for ZrB_2 and SiC of $5.4 \pm 1.6 \mu\text{m}$ and $5.3 \pm 0.5 \mu\text{m}$, respectively. The minimum density recorded was $4.97 \pm 0.02 \text{ g/cm}^3$ (95% theoretical) for the sample heated at 50°C/min, which resulted in average grain sizes for ZrB_2 and SiC of $2.5 \pm 0.7 \mu\text{m}$ and $2.7 \pm 0.3 \mu\text{m}$, respectively, less than half those obtained by sintering at 10°C/min.

Constant heating rate densification curves are shown in Figure 3A. The green density of each specimen was carefully controlled at $2.70 \pm 0.02 \text{ g/cm}^3$ ($54 \pm 1\%$ theoretical) to maintain a consistent initial microstructural state. The data are well fit by sigmoid functions delineating three stages of densification. Initial stage sintering is completed by 1700–1850°C while intermediate stage sintering where most shrinkage occurs is completed by 2150–2250°C. Densification rates are shown in Figure 3B with apparent maxima occurring between 70% and 75% relative density.

Figure 4 shows an Arrhenius plot used to calculate the apparent activation energies for densification at different relative densities. The coefficients of determination (i.e., R^2 goodness-of-fit parameter) for linearized Arrhenius expressions decrease for data obtained at higher relative densities, leading to increased uncertainties in best estimates for apparent activation energies. This behavior is characteristic for this analysis approach, as specimens approaching the final sintering stage exhibit non-negligible grain growth that violates the assumption of constant grain size during densification used in deriving the linearized Arrhenius expressions.³⁷ Calculated values for apparent activation energy at a relative density of 70%–75% are taken herein as best estimates for the activation energy for densification, that is, $940 \pm 90 \text{ kJ/mol}$.

The MSC approach has also been applied to estimate activation energy of densification. The MSC for pressureless sintering of extrusion cast billets of ZrB_2 -SiC prepared in this work was calculated by plotting relative density against the “work of sintering” parameter

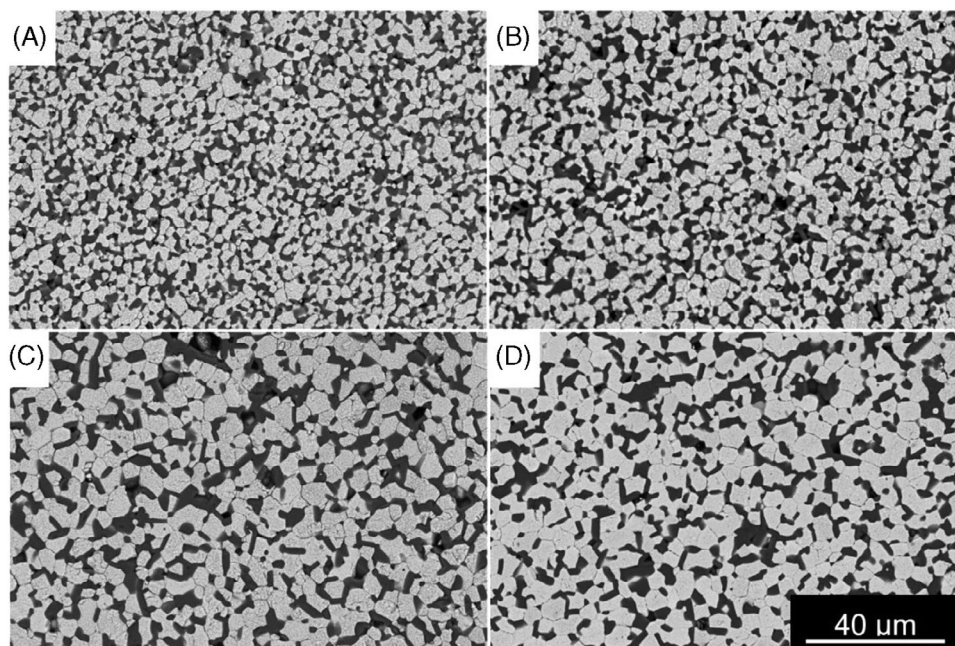


FIGURE 2 Scanning electron microscopy (SEM) micrographs of thermally etched additive manufacturing (AM)-ZrB₂-SiC heated to a maximum sintering temperature of 2300°C at different heating rates: 50°C/min (A), 35°C/min (B), 20°C/min (C), and 10°C/min (D). Light and dark phases correspond to ZrB₂ and SiC, respectively.

TABLE 1 Microstructural characteristics of ZrB₂-SiC sintered at a maximum temperature of 2300°C at different heating rates without isothermal holds. Grain size and density values are reported with standard deviations.

Heating rate, °C/min	$d(\text{ZrB}_2)$, μm	SiC major axis, μm	SiC minor axis, μm	$d(\text{SiC})$, equivalent diameter, μm	ρ , g/cm ³	Relative density (RD), %
10	5.4 ± 1.6	4.8 ± 1.3	1.5 ± 0.5	5.3 ± 0.5	5.19 ± 0.02	99 ± 1
20	4.5 ± 1.5	3.7 ± 1.2	1.3 ± 0.5	4.4 ± 0.5	5.13 ± 0.02	98 ± 2
35	3.2 ± 1.3	2.9 ± 0.8	1.1 ± 0.4	3.5 ± 0.4	5.01 ± 0.02	96 ± 2
50	2.5 ± 0.7	2.4 ± 0.6	0.8 ± 0.3	2.7 ± 0.3	4.97 ± 0.02	95 ± 2

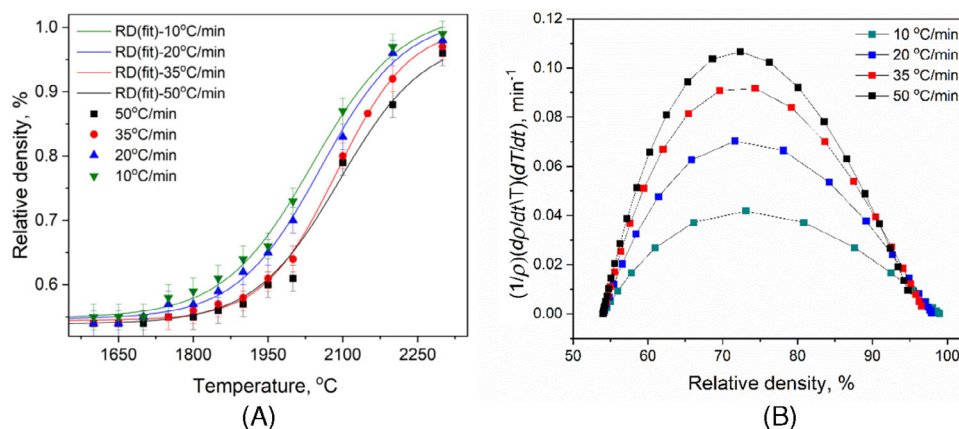


FIGURE 3 Temperature dependence of the relative density (RD) of ZrB₂-SiC (A) and densification rates as a function of RD measured at different heating rates (B).

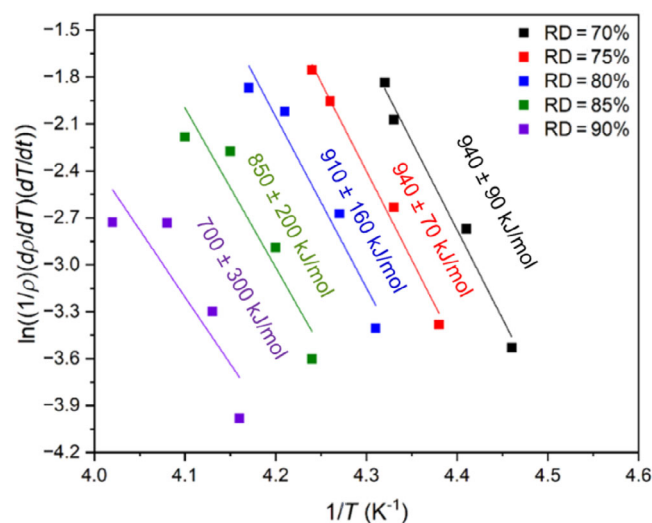


FIGURE 4 Arrhenius plots of used for calculation of apparent activation energy of densification for extrusion cast ZrB_2 -SiC.

$\Theta = \int_0^t \frac{1}{T} e^{-Q/RT} dt$ (Figure 5A) using a best fit estimate for the fitting parameter $Q = 873 \text{ kJ/mol}$ (Figure 4B) equated to the activation energy for densification. This estimate is within the standard deviation of the activation energy of $940 \pm 90 \text{ kJ/mol}$ determined by the method of Wang and Raj.

Unfortunately, data on activation energies for densification of ZrB_2 -SiC composites by pressureless sintering against which this value may be compared are scarce. Activation energies for reactive hot-press sintering of ZrB_2 -SiC reported in ref.³⁸ suggest the large values observed here are more likely to indicate that lattice diffusion is the dominant transport mechanism than grain boundary diffusion. Activation energies of comparable magnitude to those in this work have also been reported for the densification of SiC by lattice diffusion mechanism (e.g., up to $1022 \pm 122 \text{ kJ/mol}$ as reported in ref.³⁹).

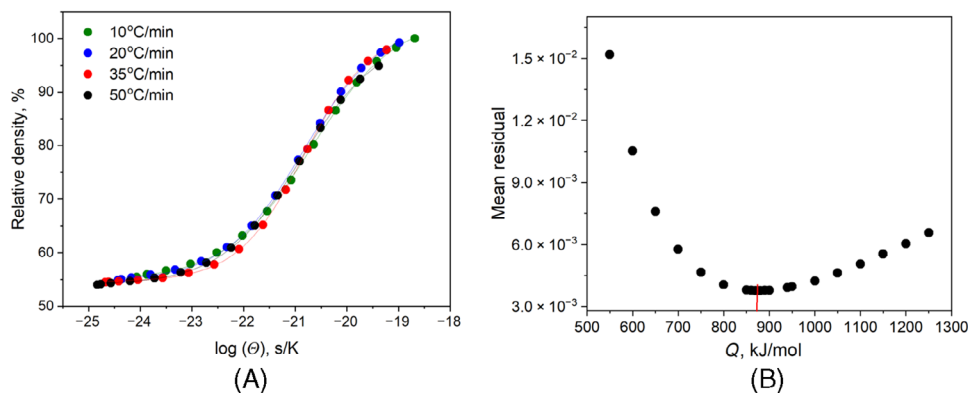


FIGURE 5 Master sintering curve (MSC) plotted for the samples densified at different heating rates (A) and mean residuals of the MSC curves at different values for Q (B).

3.2 | Grain growth behavior

The effects of sintering time during isothermal heat treatment at 2200°C on average grain sizes of ZrB_2 and SiC are illustrated in Figure 6. These SEM micrographs show changes in microstructures with light regions corresponding to ZrB_2 and dark regions corresponding to SiC. ZrB_2 grains are equiaxed under all heat treatment conditions; however, SiC grains are anisometric with plate-like morphologies whose aspect ratios vary with time-temperature heat treatment profiles (Figure 7). The SiC aspect ratio increased from 2:1 (i.e., the ratio of major to minor ellipsoidal axes averaged over all measured grains) in the initial sintered ZrB_2 -SiC specimen to approximately 3.6:1 after 24 h at 2300°C , indicating the existence of preferential growth directions.

Grain growth exhibits parabolic kinetics with normal grain growth exponent $n = 2$ as indicated by a linear fit ($R^2 > 0.97$) of normalized squared grain size versus time as shown in Figure 8. Apparent activation energies for grain growth were calculated by using Arrhenius plots shown in Figure 9 and determined to be $267 \pm 10 \text{ kJ/mol}$ for ZrB_2 and $264 \pm 15 \text{ kJ/mol}$ for SiC. No significant difference between ZrB_2 and SiC grain growth behavior was apparent, which suggests the same coarsening mechanisms may be rate-limiting. The grain growth exponent of $n = 2$ indicates diffusion control, though unambiguous determination of diffusion mechanisms cannot be ascertained with available information given a lack of literature data.

3.3 | Mechanical properties

Billets from which test specimens were obtained for the evaluation of mechanical and thermal properties were additively manufactured by computer-controlled

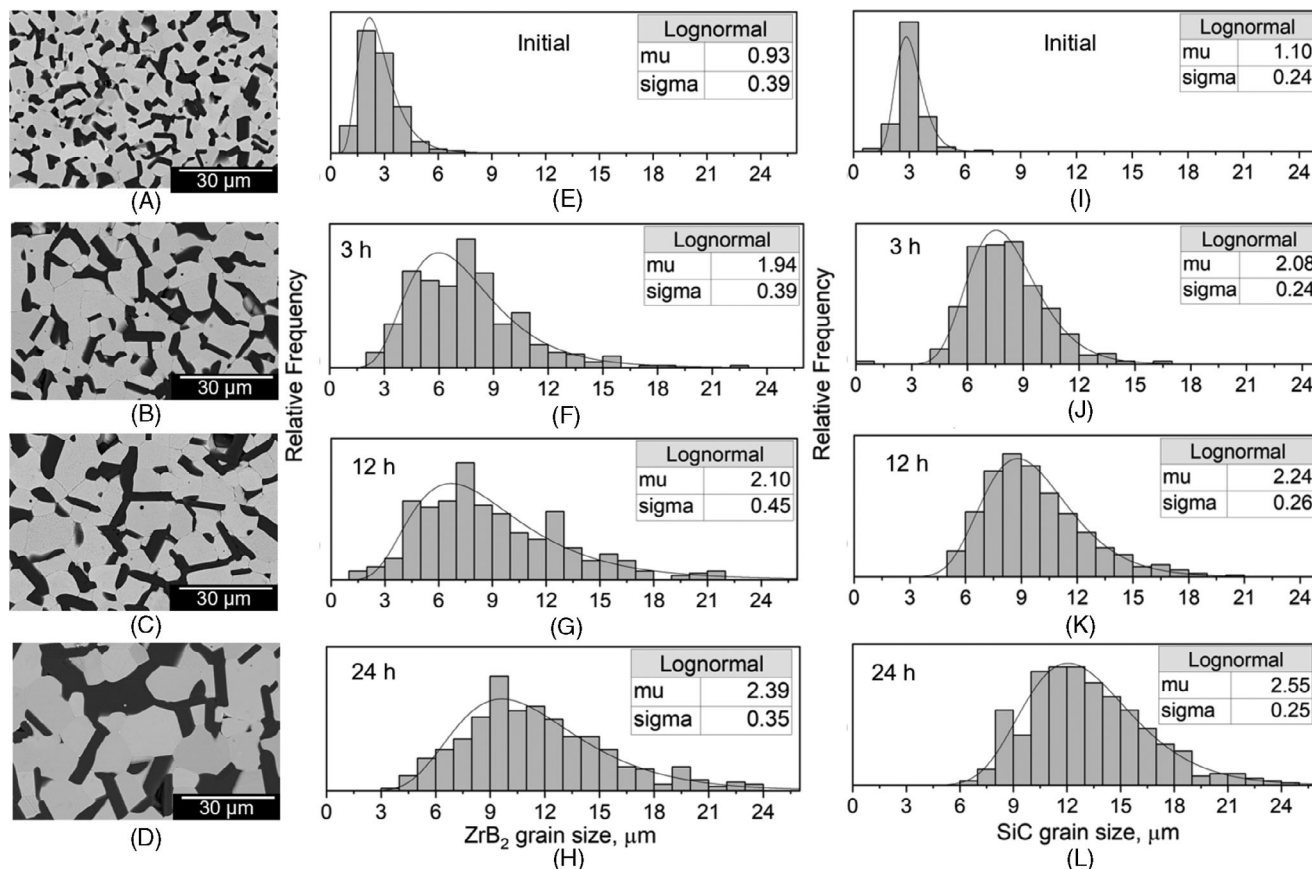


FIGURE 6 Scanning electron microscopy (SEM) images showing progressive coarsening of ZrB₂-SiC (A-D). Initial microstructure (A) and after heat treatment at 2200°C for 3, 12, and 24 h (B-D); ZrB₂ grain size distribution (E-H), SiC grain size distribution (I-L).

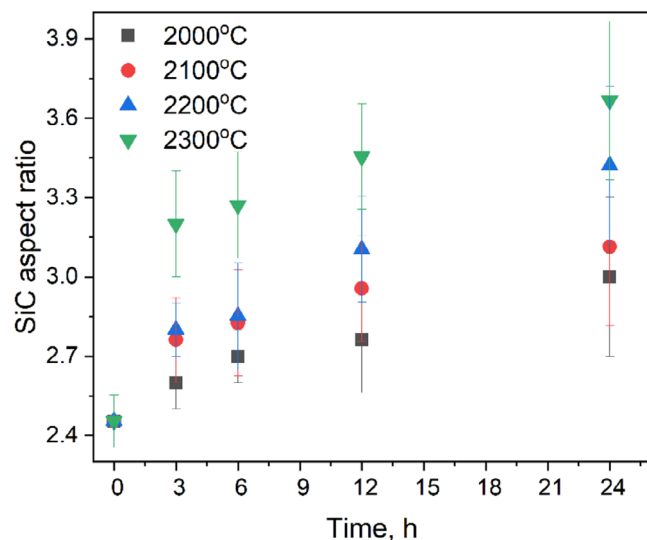


FIGURE 7 Variation in SiC aspect ratio as a function of time during isothermal coarsening trials from 2000 to 2300°C for 24 h.

layer-by-layer extrusion of high solids loading pastes (denoted herein as AM-ZrB₂-SiC). The measured density for AM-ZrB₂-SiC billets after sintering at 2100°C for

3 h was 5.92 g/cm³ (95% theoretical). Measured flexural strength, Young's modulus, and Vicker's microhardness for AM-ZrB₂-SiC are summarized in Table 2.

Young's modulus and Vickers microhardness values for AM-ZrB₂-SiC are comparable to conventionally processed materials, illustrative of the relative insensitivity of these properties to microstructure. These results are reasonable, as pressurelessly sintered ZrB₂-SiC ceramics formed by die compaction have been shown to exhibit similar Young's modulus, hardness, and fracture toughness as compared to hot-pressed materials.⁴² The lower Young's modulus of the AM-ZrB₂-SiC specimens as compared to hot-pressed specimens may reflect the presence of additional porosity.

By contrast, flexural strength in AM-ZrB₂-SiC specimens is lower by a factor of two or more as compared to hot-pressed specimens. This result is consistent with lower flexural strengths typically measured for ZrB₂-SiC ceramics densified by pressureless sintering as compared to hot-pressing due to increased flaw sizes correlated with coarser microstructural features. The lower strengths of the present materials are likely due to the presence of pores with an average size of 30–35 μm distributed uniformly throughout the microstructures of AM-ZrB₂-

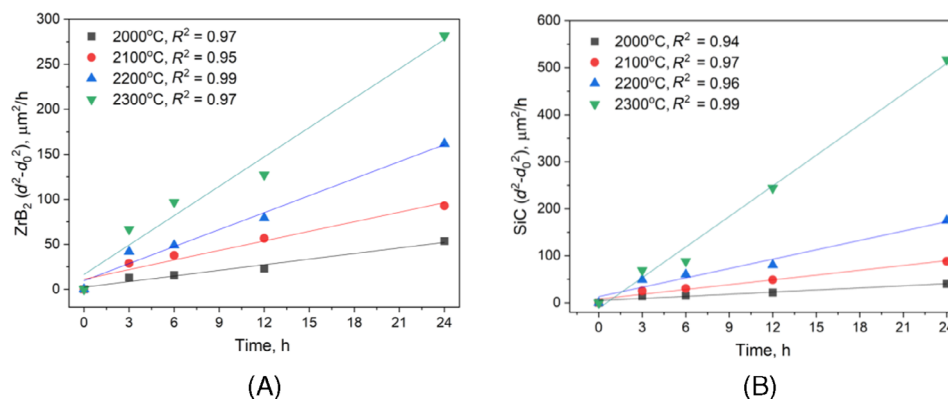


FIGURE 8 ZrB_2 (A) and SiC (B) grain sizes as functions of the time at different sintering temperatures.

TABLE 2 Comparison of mechanical properties and grain size for different manufacturing methods of $ZrB_2 + 30$ vol.% SiC.

Processing route	Relative density (RD), %	Flexural strength, MPa	Young's modulus, GPa	Vickers hardness HV0.2, GPa	$d(ZrB_2)$, μm	$d(SiC)$, μm
Hot-pressing ^{3,40,42}	99	720–1070	501–508	21.3	2.2 ± 0.5	1.2–2.7
Pressureless sintering of die compacted ^{27,41}	99	490 ± 70	490 ± 7	22	2.6 ± 0.8	2 ± 0.2
AM- ZrB_2 -SiC (this work)	95	270 ± 30	440 ± 10	20	7.3 ± 1.1	7.1 ± 0.8
AM- ZrB_2 -SiC with paste de-airing (this work)	97	270 ± 40	480 ± 20	21	7.3 ± 1.1	7.1 ± 0.8

Abbreviation: AM, additive manufacturing.

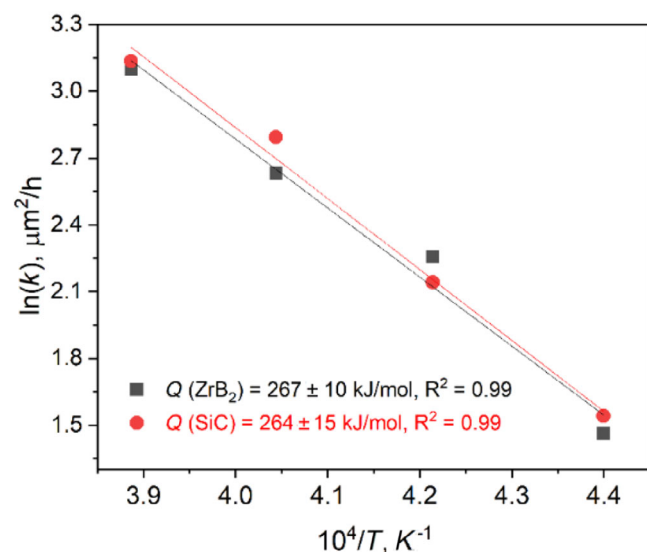


FIGURE 9 Arrhenius plots for the determination of apparent activation energy for normal growth of ZrB_2 and SiC grains during coarsening heat treatments at 2000–2300°C for 3 h.

SiC (Figure 10). These pores are not associated with inhomogeneous particle packing—indeed, the regions between pores appear to be sintered to near-full density. The pore size distribution analysis indicated that the largest pores, which measured up to 230 μm , were a result of air bubbles entrained during paste preparation. The

application of an additional vacuum de-airing stage during high shear mixing reduced the prevalence of these largest pores (Figure 10B,D), but did not eliminate them. Hence, de-airing did not increase the strength of AM- ZrB_2 -SiC in the present study.

After de-airing, the maximum observed pore size was 153 μm while the mean pore size exhibited a marginal decrease from 35 to 29 μm . The removal of large air bubbles contributed to an increased bulk density of 5.05 g/cm³ (97% theoretical). The decreased porosity in the de-aired sample is likely responsible for the observed increase in Young's modulus shown in Table 2.

Pores not associated with air bubbles are thought to originate from gelation artifacts in the pastes used to AM test billets in this work. The cause of these gelation artifacts is not well understood, though the total porosity directly correlates with the batched amount of binder and dispersants.¹⁷ Increases in flexural strength and sintered density will require the elimination of these pores by improving paste formulations.

3.4 | Thermal properties

Thermal diffusivities obtained from laser flash can be seen in Figure 11. Thermal diffusivity decreases with increasing temperature from 35.0 mm²/s at 298 K (25°C) down

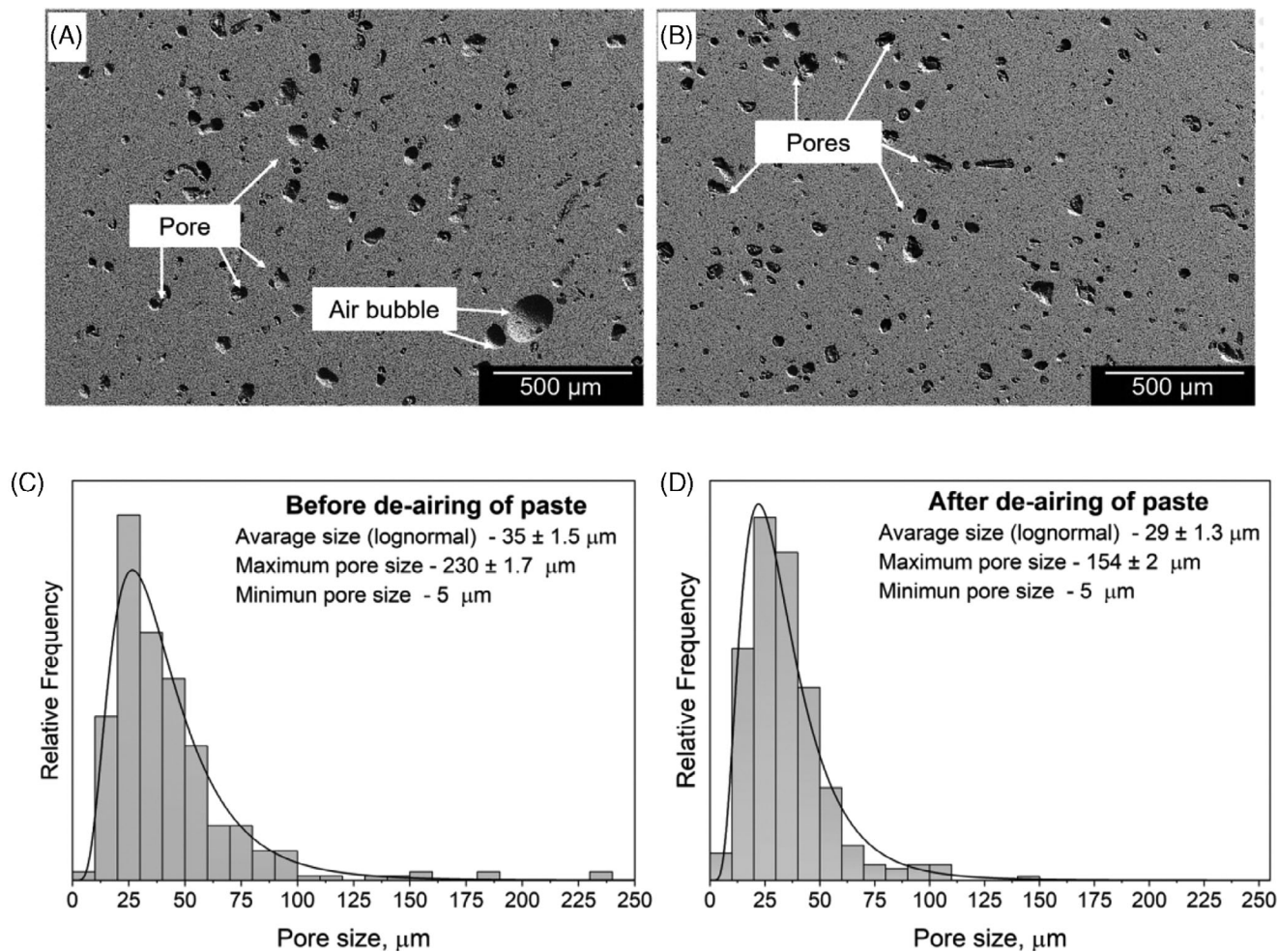


FIGURE 10 Scanning electron microscopy (SEM) micrographs of additive manufacturing (AM)- ZrB_2 -SiC sintered at 2100°C with a heating rate of $20^\circ\text{C}/\text{min}$ with 3-h isothermal hold, both before (A) and after (B) the introduction of additional paste de-airing stages. Results of the pore size distribution analysis for samples before (C) and after (D) the de-airing process.

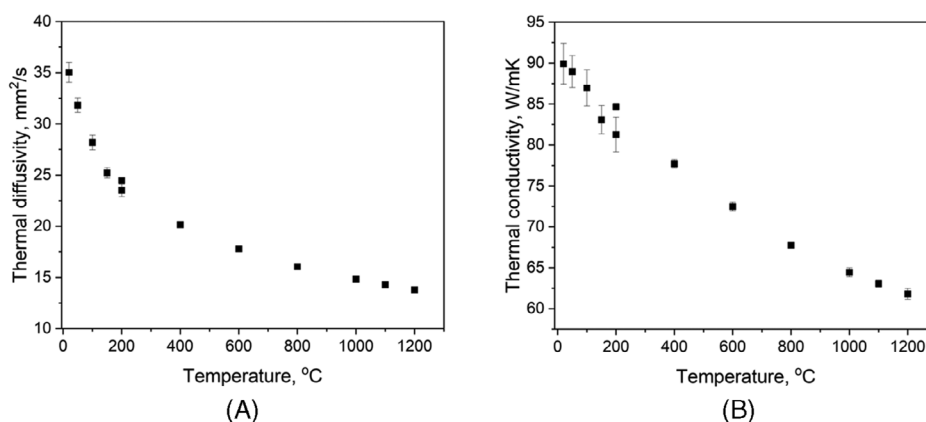


FIGURE 11 Thermal diffusivities (A) and thermal conductivities (B) determined for additive manufacturing (AM)- ZrB_2 -SiC samples as functions of temperature.

to 13.8 mm²/s at 1473 K (1200°C). This behavior corresponds to the decrease in the phonon contribution to thermal conductivity as reported in⁴ as well as the increase in electrical resistivity both decreasing thermal diffusivity. Thermal conductivities calculated using the measured thermal diffusivities followed the same trend with increasing temperature decreasing from 89.9 W/m K at 298 K (25°C) to 61.8 W/m K at 1473 K (1200°C). Both values are close to what is expected for nominally pure ZrB₂-based ceramics.⁴³ This work collected thermal diffusivity and density values for additively manufactured samples but utilized tabulated values of thermal expansion and heat capacity from ref.³⁴ for nominally pure ZrB₂ and SiC along with a rule of mixtures approximation. This approximation likely overestimates the thermal conductivity as it fails to consider the decrease in phonon conduction caused by the presence of multiple phases and large pore defects. However, despite these differences, the overall trends of decreasing thermal conductivity with increasing temperature were similar.

4 | CONCLUSIONS

Microstructure evolution, densification and grain growth behaviors, mechanical and thermal properties of additively manufactured ZrB₂-SiC were investigated in this work. The following conclusions can be drawn:

The densification behavior of additively manufactured ZrB₂-SiC ceramics is significantly influenced by heating rates during pressureless sintering. The maximum relative density achieved was 99.9% at a heating rate of 10°C/min to a maximum sintering temperature of 2300°C, while the minimum was 95.0% at 50°C/min to 2300°C without isothermal hold.

Best estimates for the apparent activation energy for densification were determined to be 940 ± 90 kJ/mol using the method of Wang and Raj and 873 kJ/mol using the MSC approach, suggesting that lattice diffusion is the rate controlling densification mechanism although further densification studies systematically varying compositions and microstructures appear warranted to more fully understand why the magnitude of these apparent activation energies are so large.

Normal grain growth with grain growth exponent $n = 2$ is indicated for ZrB₂ and SiC, indicating diffusion-controlled coarsening. The activation energies for grain growth were 267 ± 10 for ZrB₂ and 264 ± 15 for SiC, implying the same coarsening mechanisms.

Additively manufactured ZrB₂-SiC composites exhibited homogeneous microstructures but with elevated porosity arising from paste gelation artifacts, resulting in a maximum relative density of 95% upon sintering

at 2100°C for 3 h at 30°C/min heating rate. The presence of these pores reduced the flexural strength by a factor of two or more as compared to conventionally processed ZrB₂-SiC, although Young's modulus, Vicker's microhardness, and thermal diffusivity/conductivity values did not exhibit significant knockdowns in measured property values as compared to conventionally sintered and hot-pressed counterparts.

This study provides valuable insights into the processing-microstructure-property relationships of additively manufactured ZrB₂-SiC composites and highlights the challenges and opportunities for further optimization of their performance through process control and microstructure engineering.

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